



Map Problem Bank — HashMap, LinkedHashMap, TreeMap (Java)

Legend

★ = Core Placement Problem (Mandatory)

Covers internal implementation details alongside application-level problems.

These problems should be completed by every student before moving to Queue & Deque.

Section 1: Map Basics and Creation

★ Create and Display a HashMap

★ Create and Display a LinkedHashMap

★ Create and Display a TreeMap

★ Add Key-Value Pairs using put()

★ Access Value using get()

Access Value with a Default using getOrDefault()

Find Size of a Map using size()

Create an Immutable Map using Map.of()

Section 2: HashMap vs LinkedHashMap vs TreeMap Internals (Core Concept)

★ Internal Structure of HashMap (Buckets, Hashing, Collision Handling)

★ Internal Structure of LinkedHashMap (Hash Table + Doubly Linked List for Order)

★ Internal Structure of TreeMap (Red-Black Tree, Sorted by Key)

★ Demonstrate Key Ordering Differences: HashMap vs LinkedHashMap vs TreeMap

★ HashMap Collision Handling and Treeification (Java 8+ Bucket-to-Tree)

Load Factor and Resizing Behavior in HashMap

★ Custom Object as Key — hashCode()/equals() Contract

★ Time Complexity Comparison: get()/put() — HashMap vs TreeMap

Section 3: Adding, Updating, and Removing

★ Update Value of an Existing Key

★ Remove a Key-Value Pair using remove()

Insert Only if Key is Absent using putIfAbsent()

★ Combine Values using merge()

Compute and Update Value using compute() / computeIfAbsent()

Clear All Elements from a Map

Section 4: Searching and Checking

★ Check if a Key Exists using containsKey()



★ **Check if a Value Exists using containsValue()**

★ **Find the Key with the Maximum Value**

Find the Key with the Minimum Value

Section 5: Iterating and Viewing

★ **Iterate through Keys using keySet()**

★ **Iterate through Values using values()**

★ **Iterate through Entries using entrySet()**

Iterate using forEach() with Lambda

Get All Keys as a List

Get All Values as a List

Section 6: Sorting and Ordering

★ **Sort a HashMap by Keys**

★ **Sort a HashMap by Values**

★ **Sort Map Entries using Streams and Comparator**

Convert Sorted Map Entries into a LinkedHashMap

Section 7: Map-Based Logical Problems

★ **Count Frequency of Words in a Sentence**

★ **Count Frequency of Characters in a String**

Find Common Keys between Two Maps

★ **Merge Two Maps Summing Values of Common Keys**

★ **Group a List of Objects into a Map (Grouping by Property)**

Invert a Map (Swap Keys and Values)

Find Duplicate Values in a Map

Section 8: Real World Map Applications

★ **Student Record Management using Map**

★ **Employee Salary Lookup System**

Inventory Management using Map

★ **Phone Book / Contact Directory using Map**

★ **Login Credential Validator using Map**

Word Count Analyzer for a Document using Map